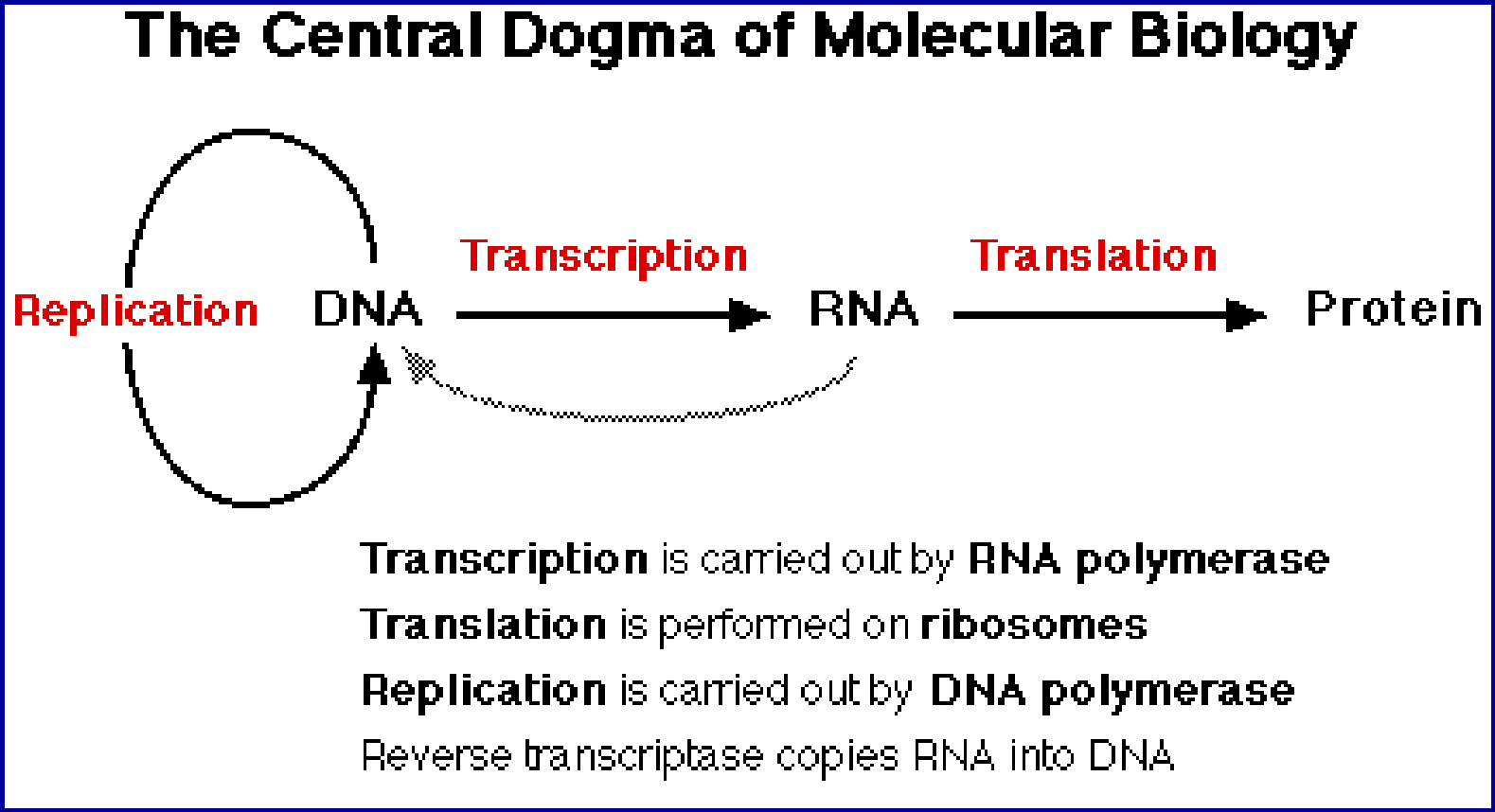
Translation (Protein Synthesis)

RNA → protein

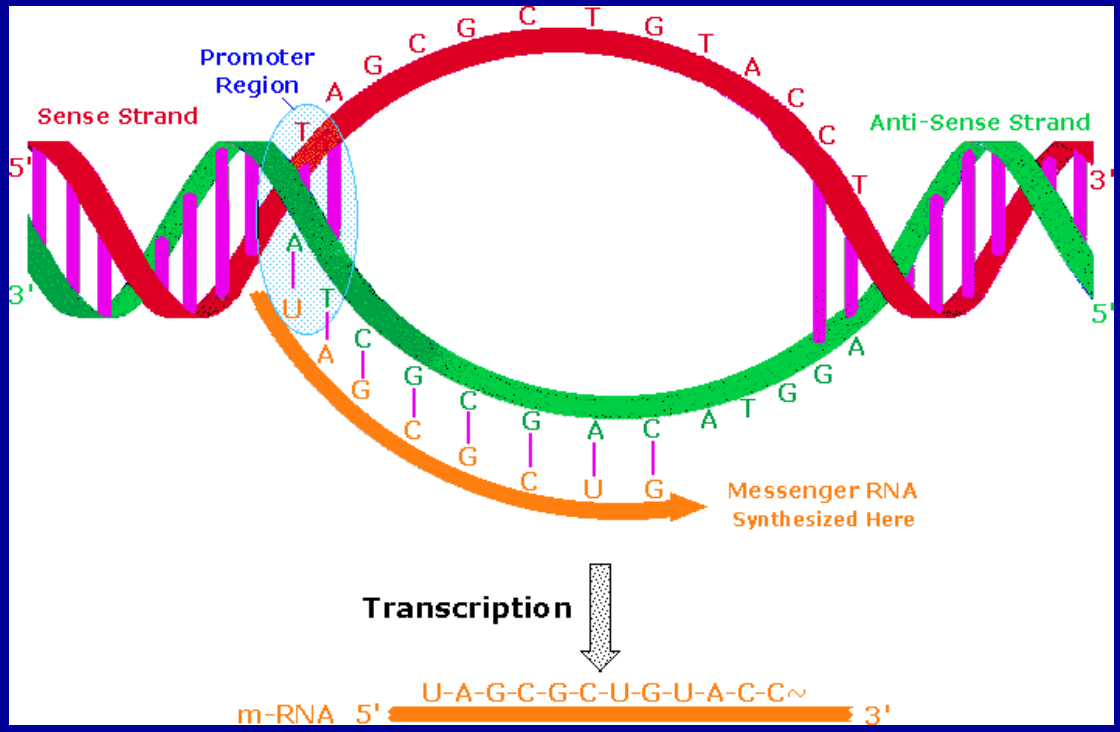


Making a protein

- Many RNAs needed
  - mRNA, tRNA, rRNA

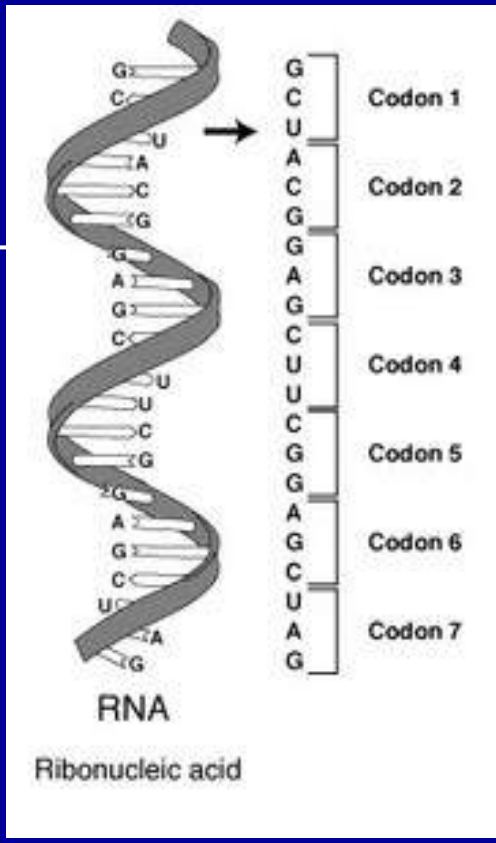
Messenger RNA (mRNA)

- Carries coded instructions for protein synthesis (translation)
  - From the DNA in the nucleus to the ribosome



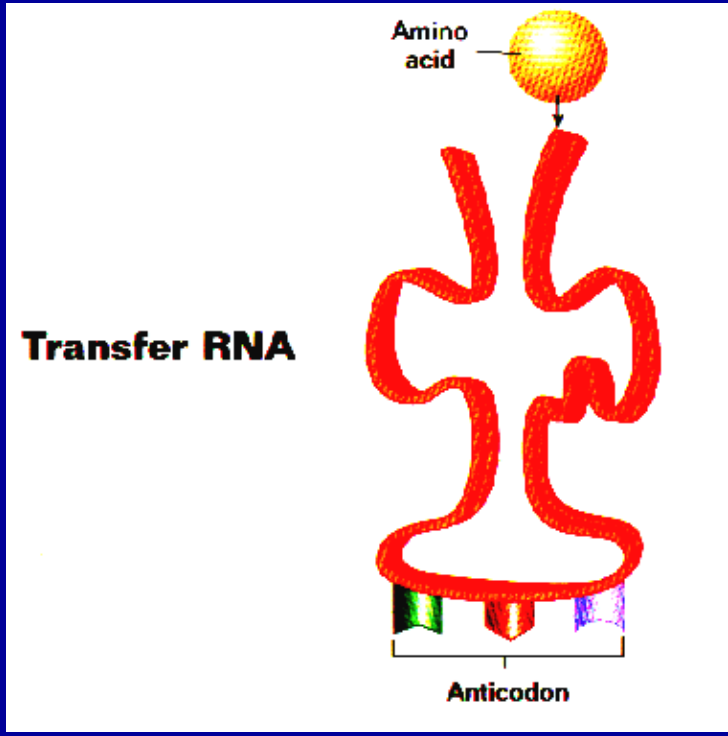
The genetic code

- Codons
  - instructions for making a protein, a series of three nucleotides on the mRNA
  - Each codon signifies start, stop, or an amino acid



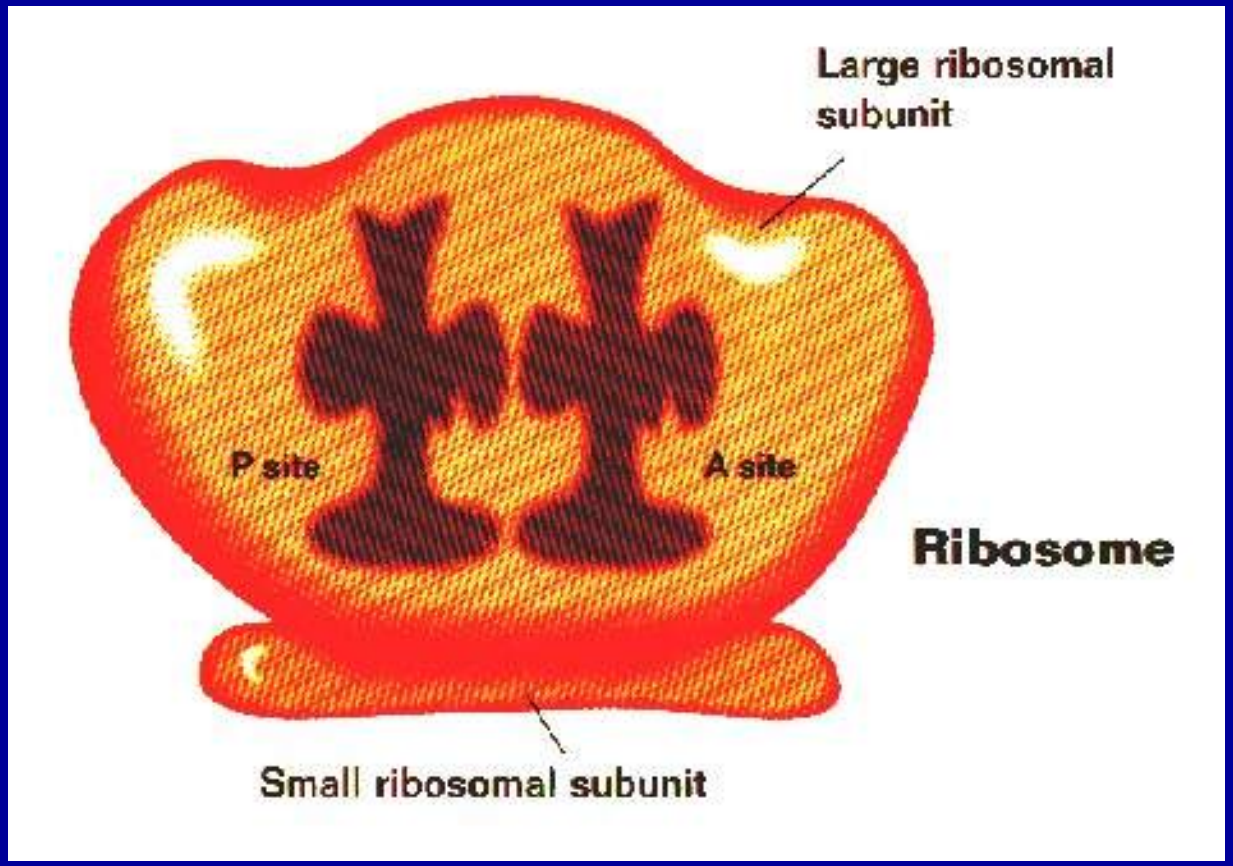
Transfer RNA (tRNA)

- Brings amino acids to the ribosome so it can build proteins
- It has Anticodons
  - 3 nucleotide sequence complementary to the mRNA codon



Ribosomal RNA (rRNA)

- Makes up ribosomes



Building a Polypeptide

- The three stages of translation:
  - Initiation
  - Elongation
  - Termination

•All three stages require protein translation factors

Ribosome Association and Initiation of Translation

1. Small ribosomal subunit binds
   mRNA and special initiator tRNA (met-tRNAi)
   (carries the amino acid methionine)

2. Small subunit scans along the mRNA until first start codon (AUG).

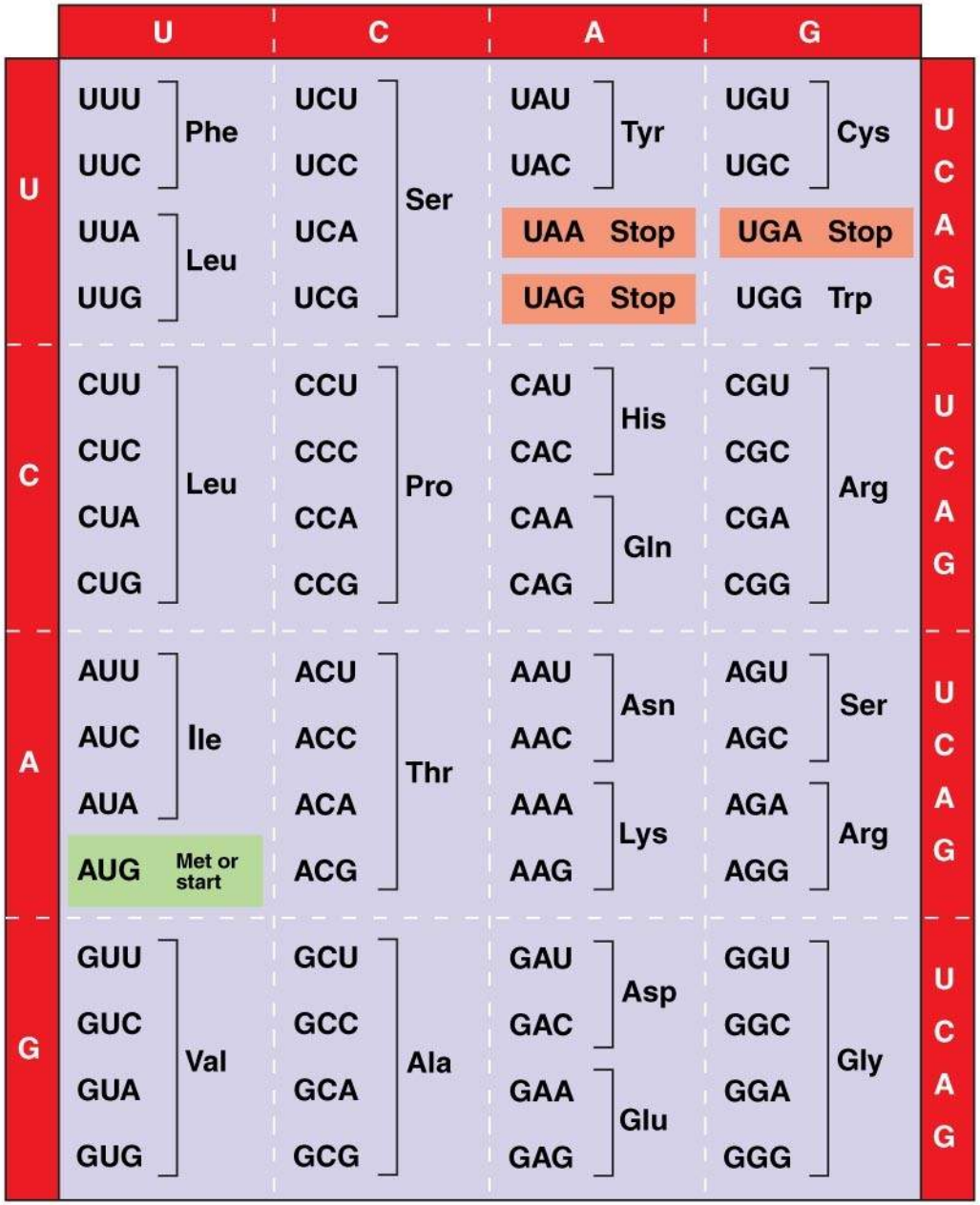
3. Initiation factors bring in large subunit
   initiator tRNA occupies the P site.

LE 17-5

Genetic Code

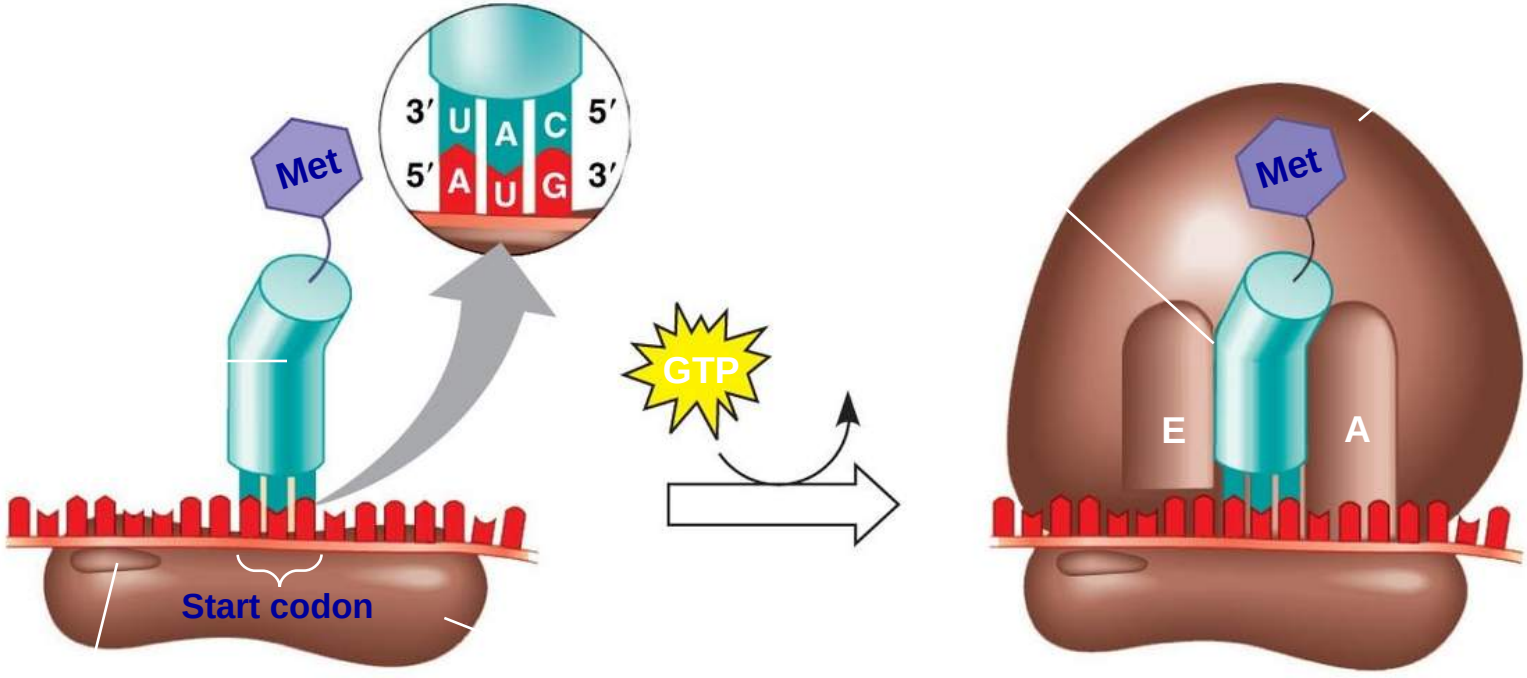
Memorize Start Codon

Codon Table



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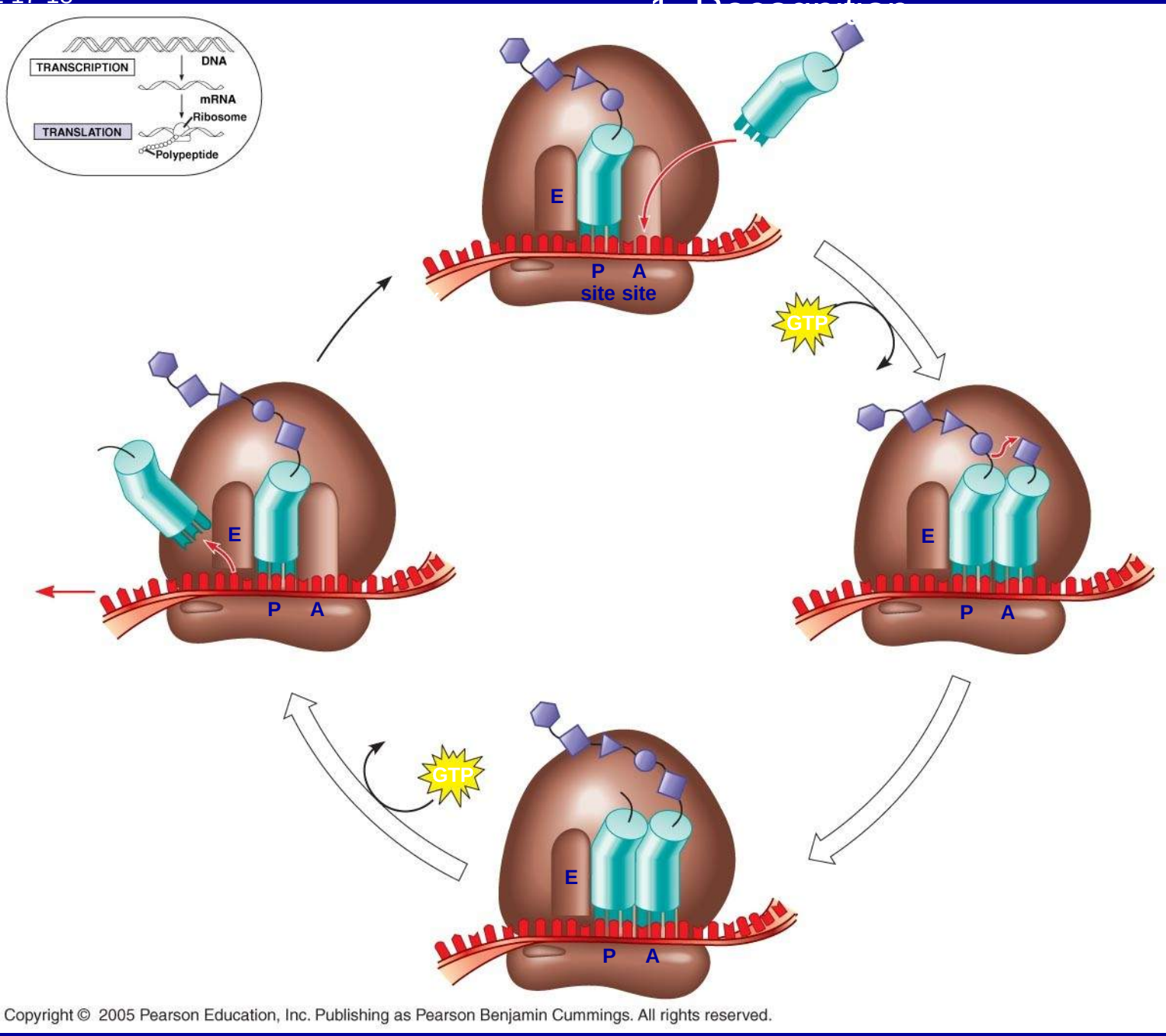
Elongation of the Polypeptide Chain

- Amino acids are added one by one to the preceding amino acid

-Elongation factors facilitate

- codon recognition
- peptide bond formation
- translocation

LE 17-18



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ond

Termination of Translation

- Occurs when stop codon in mRNA reaches A site of ribosome

- A site accepts protein called release factor

-Release factor causes addition of water molecule instead of amino acid

- Polypeptide released, ribosomal subunits dissociate and fall off mRNA

Do we need new mRNA every time?

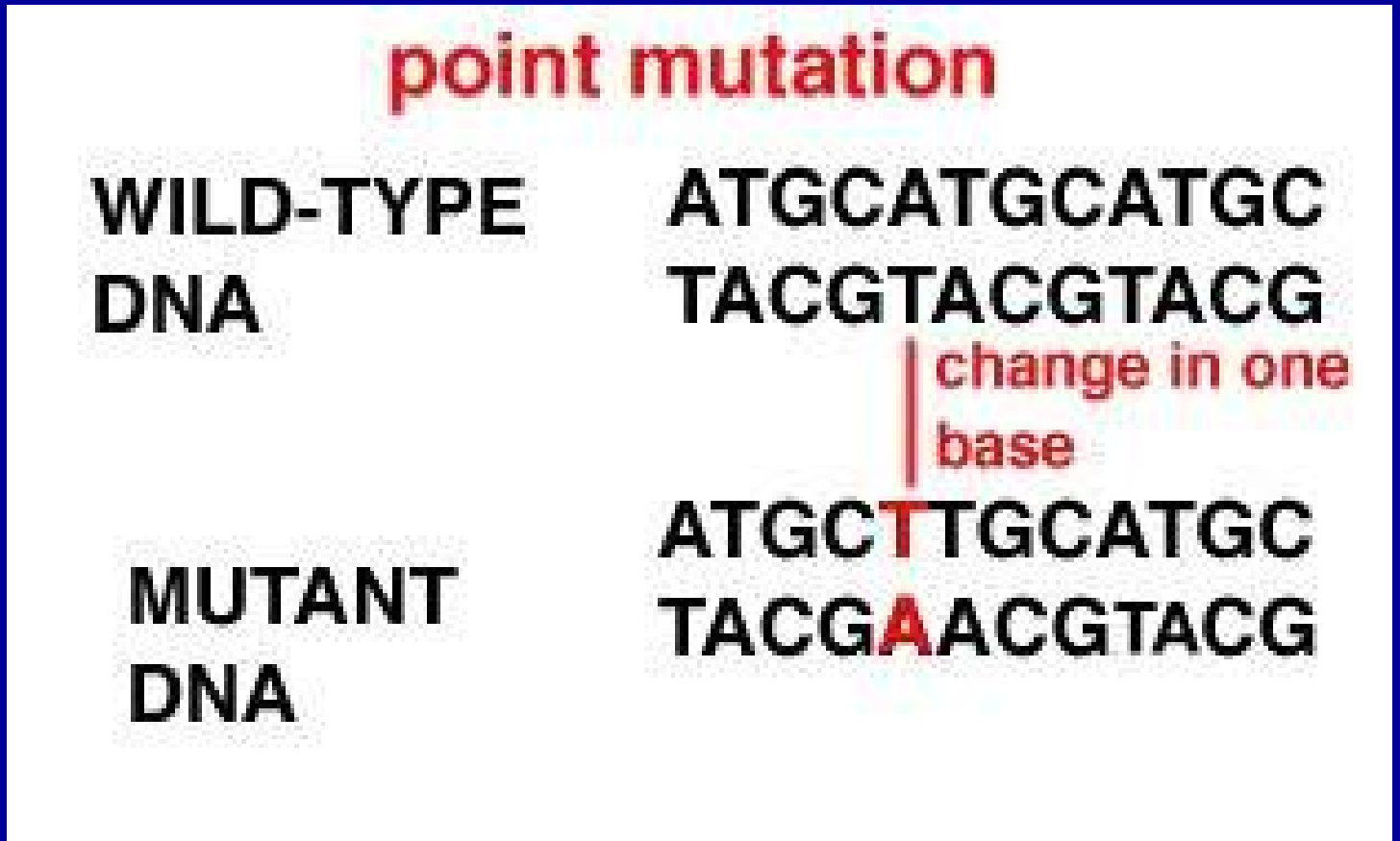
- NO

- The same mRNA may be used to make many copies of the same protein.

- http://lab.concord.org/embeddable.html#interactives/sam/DNA-to-proteins/1-dna-to-protein.json

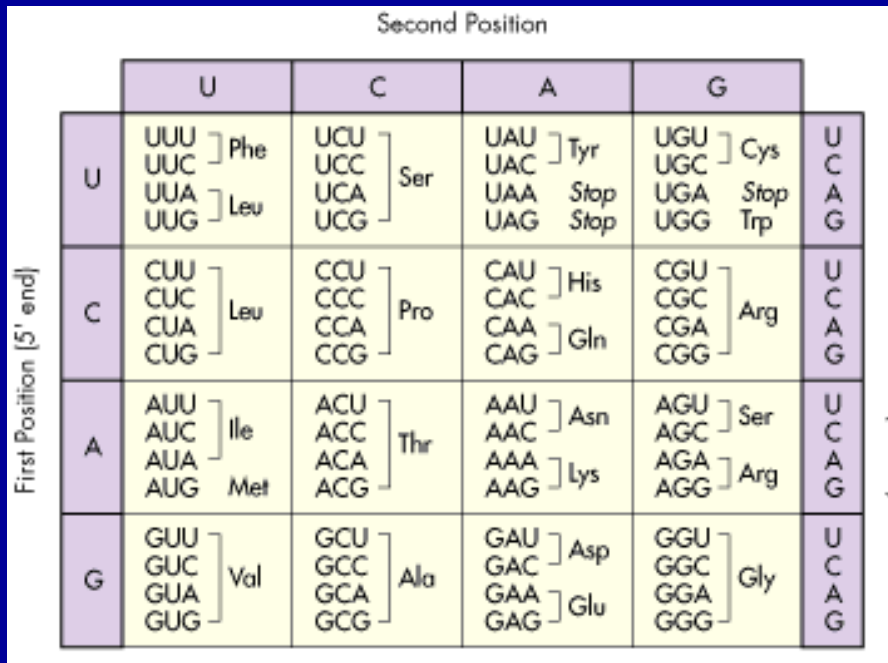
Mutations

- Point mutation
  - Change in one or a few nucleotides



Mutations

- Substitution
  - can have no effect
  - Protein could not be made
  - Protein made w/ wrong amino acid (nonfunctional protein)



First Position (5' end)	Second Position: U	Second Position: C	Second Position: A	Second Position: G	Third Position
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Mutations

- Insertions/deletions
  - 1 or more nucleotides added or deleted
  - Can result in frame shift mutations